

Decomposition-aware global optimization: certified coordinate grids, conditional recourse, and structural limits

Abstract

Sparse factorization can make global optimization useful without making every sparse problem easy. This report develops a complete certified grid algorithm on supplied tree decompositions. A unary interpolation correction makes finite-state dynamic programming a valid continuous lower bound; coordinate min-marginals then remove regions that cannot improve a feasible incumbent. Under global quadratic growth, the surviving geometric grids have state counts independent of the requested accuracy. For rational mixed-integer box quadratics this gives an exact algorithm with running time $f(p, \kappa) \text{poly}(I)$, where p is maximum bag size, I is binary input length, and κ is upper coordinate curvature divided by growth. The growth constant is not required as input, and certificate validity does not depend on it. Stationary-face recovery also gives finite exact output for arbitrary nonunique rational box quadratics, without a favorable general complexity bound. The approximation result extends to explicitly encoded fixed-degree polynomials, with different exact-output obligations. We separate these proved guarantees from implementation evidence and from extensions that require additional recourse, geometric, or certification structure. The report is a technical synthesis of repository results; priority and competitive general-purpose performance are not asserted.

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1 The solver capability and its limits

The main capability is a global bound for a factorized objective on a product domain, computed by finite tables that agree exactly on shared coordinates. It can be used as a complete solver for that domain or as a component of a larger constrained solver. The relevant parameter is *maximum bag size and conditioning together*. Small graph width alone is not the claim. Large tables, weak growth, or a poor decomposition can make the algorithm impractical even when the theorem applies.

The dependency chain starts with coordinate curvature, which yields a corrected-grid lower bound and safe min-marginal filtering. Global growth then gives localization, bounded state counts, and a quantitative output guarantee. The first two implications hold without uniqueness or growth. Growth is used to bound work, not to justify the answer. Rational arithmetic and quadratic height bounds convert the final implication into an exact mixed-integer quadratic algorithm. Polynomial objectives require their own output argument.

This distinction also places the result relative to earlier repository work. The September 29 affine-message existence bounds were centered at an optimizer and left an algorithmic issue on branching decompositions. Shared-coordinate grids subsequently supplied an optimizer-independent algorithm, and min-marginal filtering removed the accuracy exponent from its state bound. This does not prove that an older overlapping-cell affine-message representation has the same properties. In particular, counterexamples based on separator-copy drift or accumulated interpolation error still apply to the representations they concern.

The current external comparison also sets a sharp boundary. Del Pia and Khajavirad give an exact $O(n^2)$ algorithm on forests, but prove strong NP-hardness of continuous box QP at treewidth two, even with bounded coefficients; quartic path problems are also strongly NP-hard [7]. The forest result needs no growth promise. The conditioned theorem here therefore complements that work and does not dominate it. Bienstock and Muñoz treat broader constrained polynomial models

with scaled feasibility and objective tolerances [3]; the constrained extension below obtains exact feasibility under narrower integral structure.

2 Model, decomposition, and arithmetic

Let

$$\min_{x \in X} F(x), \quad F(x) = \sum_{a \in \mathcal{A}} f_a(x_{S_a}), \quad X = \prod_{i=1}^n X_i, \quad (1)$$

where $X_i = [a_i, b_i]$ or $[a_i, b_i] \cap \mathbb{Z}$. Integer endpoints are rounded inward, empty domains rejected, and fixed coordinates substituted before the construction. The zero-dimensional case is immediate. Thereafter $n \geq 1$ and $s = \max_i(b_i - a_i) > 0$.

A supplied tree T has bags B_t , maximum size p , and N nodes. Every factor scope is contained in a bag to which the factor is assigned once, and the bags containing any coordinate form a connected subtree. The decomposition and assignments can be checked directly. The input length I includes these data. A variable can occur in arbitrarily many bags; no occurrence bound is assumed.

A valid upper coordinate-curvature bound is $L > 0$ such that

$$t \mapsto F(x_1, \dots, t, \dots, x_n) - \frac{L}{2}t^2 \quad \text{is concave on each coordinate interval.} \quad (2)$$

The condition is on the full continuous hull, including fractional points between integer labels, and on the *sum* of all factors. For a quadratic with Hessian H , $L = \max_i \max\{H_{ii}, 0\}$ is valid when positive. If every diagonal is nonpositive, successive endpoint replacement does not increase the objective. Endpoint dynamic programming is then exact, with at most two labels per coordinate, without a growth assumption.

The quantitative analysis assumes a unique optimizer x^* and

$$F(x) - f^* \geq g\|x - x^*\|^2 \quad (x \in X), \quad f^* = F(x^*), \quad \kappa = \max\{1, L/g\}. \quad (3)$$

This is global growth in the original Euclidean coordinates. Near-tied integer assignments may make g small even if the continuous extension is strongly convex. Rescaling coordinates can change L/g and is not a free invariance of the theorem.

The lower-bound arguments first use exact factor evaluations and exact arithmetic. The bit-complexity theorem applies to explicit rational quadratics; Section 5.2 states its polynomial extension. An arbitrary real-valued oracle does not supply a bit-complexity theorem.

3 A corrected grid is a continuous certificate

Work in a current product subbox $B \subseteq X$. For each coordinate take a finite feasible grid G_i containing both current endpoints. At a node $v \in G_i$, let $\ell_i(v)$ be the largest adjacent interval length. For integer coordinates ignore intervals of length one; set $\ell_i(v) = 0$ if no other adjacent interval remains. Define

$$d_i(v) = \frac{L}{8}\ell_i(v)^2, \quad D(y) = \sum_i d_i(y_i), \quad Q(y) = F(y) - D(y), \quad b = \min_{y \in \prod_i G_i} Q(y). \quad (4)$$

Assign each unary correction to one containing bag. This changes no factor scope or graph edge.

Lemma 3.1 (Interpolation and conditional interpolation). *For every $x \in B$, there is a distribution on grid assignments Y with $\mathbb{E}Y = x$ and $\mathbb{E}Q(Y) \leq F(x)$. Thus $b \leq \min_B F$. Writing*

$$m_i(v) = \min\{Q(y) : y_i = v, y \in \prod_k G_k\},$$

every adjacent pair $a, b' \in G_i$ satisfies

$$\min\{m_i(a), m_i(b')\} \leq F(x) \quad \text{whenever } x \in B, x_i \in [a, b']. \quad (5)$$

If y minimizes Q , then $F(y) - b = D(y)$.

Proof. Independently round each coordinate of x to the endpoints of its containing grid interval with probabilities preserving its mean. A grid coordinate remains fixed. If an interval has endpoints u, v , its rounding variance is $(x_i - u)(v - x_i) \leq (v - u)^2/4$. Concavity in (2) gives, conditionally on all other coordinates,

$$\mathbb{E}F(x \text{ with coordinate } i \text{ rounded}) \leq F(x) + \frac{L}{2} \text{Var}(Y_i).$$

Successive independent rounds keep these variances unchanged, so their errors add. For a genuinely rounded interval of length Δ , both endpoint corrections are at least $L\Delta^2/8$, which dominates $L \text{Var}(Y_i)/2$. Integer unit intervals have no feasible interior point, so omitting them causes no missing rounding error. Coordinates already on their grids have zero variance and nonnegative correction. Subtracting the corrections proves $\mathbb{E}Q(Y) \leq F(x)$. The minimum is no larger than this expectation. If $x_i \in [a, b']$, the same rounding has $Y_i \in \{a, b'\}$, so its expected corrected value is at least the left side of (5). The last equality is the definition of b at its minimizing assignment. $\square$

3.1 One tree computation supplies all coordinate bounds

Let q_t be the sum of factors and negative unary corrections assigned to bag B_t . For an oriented tree edge $t \rightarrow u$, let $S_{tu} = B_t \cap B_u$. The message recurrence is

$$M_{t \rightarrow u}(x_{S_{tu}}) = \min_{x_{B_t \setminus S_{tu}}} \left[q_t(x_{B_t}) + \sum_{v \in \mathcal{N}(t) \setminus \{u\}} M_{v \rightarrow t}(x_{B_v \cap B_t}) \right]. \quad (6)$$

All variables range over their coordinate grids. Empty separators have one scalar state. Two tree passes compute both orientations. Set

$$A_t(x_{B_t}) = q_t(x_{B_t}) + \sum_{v \in \mathcal{N}(t)} M_{v \rightarrow t}(x_{B_v \cap B_t}). \quad (7)$$

Induction on a cut tree edge proves that $M_{t \rightarrow u}$ minimizes exactly the factors in the t component for the fixed separator labels. Running intersection makes the remaining component variables disjoint once those labels are fixed. Hence A_t is the exact global corrected min-marginal at bag t . Its minimum is b , and minimizing it with $x_i = v$ gives $m_i(v)$ for any containing bag. Stored argmin choices recover a global minimizer.

If $K = \max_i |G_i|$, all messages and coordinate min-marginals require

$$O(p(N + |\mathcal{A}|)K^p) \quad (8)$$

table operations, plus factor evaluation and index costs. Incoming messages are accumulated by sums, and outgoing sums use exclusions; there is no product over the children of a bag. This is the familiar finite-state junction-tree computation, used here for an explicit continuous lower bound. Variable elimination, tree agreement and reparameterization are established tools; see, for example, [6, 13].

3.2 Filtering and the certificate for the original domain

Maintain a feasible point realizing a nonincreasing upper bound U . After finding $y \in \arg \min Q$, replace U by $\min\{U, F(y)\}$. Retain a coordinate interval $[a, b']$ exactly when

$$\min\{m_i(a), m_i(b')\} \leq U. \quad (9)$$

Replace each coordinate domain by the hull of its retained intervals; a singleton stays a singleton. Use y as the next center.

Proposition 3.2 (Safe filtering history). *Every global optimizer, the incumbent, and y survive the filter. The last grid bound, together with all preceding filtering records, is a valid lower bound on the original box. This assertion uses neither growth nor uniqueness.*

Proof. The conditional bound (5) makes every interval containing a coordinate of a global optimizer or the incumbent pass (9). Also $m_i(y_i) \leq Q(y) = b \leq f^* \leq U$, so every coordinate of y survives. Taking hulls can only reintroduce points. Any point removed from the original box has a first stage at which one of its coordinates is excluded. At that stage its containing discarded interval has conditional lower bound greater than the stage's U . Its objective is therefore greater than the final nonincreasing U . The last grid lower bound applies to all surviving points, and is at most $f^* \leq U$. Together the two statements bound every original point. Keeping only the final restricted grid would omit the first part of this argument and is insufficient. $\square$

The certificate records input factors, curvature evidence, grids, corrections, directed messages or enough data to recompute them, coordinate filtering decisions, and feasible upper bounds. A checker can recompute the finite minima exactly. Growth estimates influence neither the checking rule nor its validity.

4 Why filtering removes the accuracy exponent

At stage j use $h_j = s2^{-j}$ and a center c in the current box. For a fixed $0 < \theta \leq 1/4$, build outward from c_i . At distance t take a continuous step $h_j + \theta t$ or an integer step $\max\{1, \lfloor h_j + \theta t \rfloor\}$, clipping at an endpoint. The relevant adjacent length satisfies

$$\ell_i(v) \leq h_j + \theta |v - c_i|. \quad (10)$$

For an integer interval of length at least two this follows from the floor step; unit intervals have zero correction. Clipping only decreases the length. Thus for every grid assignment z ,

$$D(z) \leq \frac{L}{4}nh_j^2 + \frac{L\theta^2}{2} \left(\|z - x^*\|^2 + \|c - x^*\|^2 \right). \quad (11)$$

Lemma 4.1 (Contraction). *Suppose $\theta^2 \leq 1/(8\kappa)$. Put $B_0 = \max\{1, 4L/(11g)\} \leq \kappa$. If y_j is the stage's corrected minimizer, and the initial center is any feasible point, then*

$$E_j := \|y_j - x^*\|^2 \leq \frac{4L}{15g}nh_j^2 + \frac{E_{j-1}}{15}, \quad (12)$$

$$E_j \leq B_0nh_j^2, \quad F(y_j) - b_j = D(y_j) \leq \frac{7L}{8}nh_j^2. \quad (13)$$

Here E_{-1} denotes the initial center's squared error.

Proof. The restricted box still contains x^* . Lemma 3.1 and growth give $gE_j \leq F(y_j) - f^* \leq D(y_j)$. Since $L\theta^2/2 \leq g/16$, absorbing the first distance term in (11) proves (12). Initially $E_{-1} \leq ns^2$, and $4L/(15g) + 1/15 \leq B_0$. At later stages $h_{j-1} = 2h_j$, so induction uses $4L/(15g) + 4B_0/15 \leq B_0$. At all stages $E_{j-1} \leq 4B_0nh_j^2$. Substitution into (11) gives

$$D(y_j) \leq \frac{L}{4}(1 + 10\theta^2 B_0)nh_j^2 \leq \frac{7L}{8}nh_j^2,$$

where $\theta^2 B_0 \leq 1/4$ is a deliberately loose bound. Filtering does not alter any step of this proof, because it retains the optimizer and the next center. $\square$

Lemma 4.2 (Localization of retained intervals). *After stage j , the retained continuous domain in coordinate i lies within distance $5\sqrt{n\kappa}h_j$ of $(y_j)_i$. The integer domain lies within distance $1 + 5\sqrt{n\kappa}h_j$.*

Proof. An interval passing (9) has an endpoint v and a complete grid assignment z with $z_i = v$ and $Q(z) = m_i(v) \leq U_j$. By (13), $U_j - f^* \leq 7Lnh_j^2/8$. Consequently

$$g\|z - x^*\|^2 \leq \frac{9L}{8}nh_j^2 + \frac{L\theta^2}{2}(\|z - x^*\|^2 + 4B_0nh_j^2).$$

Absorb as before to obtain

$$\|z - x^*\|^2 \leq \left(\frac{6L}{5g} + \frac{4B_0}{15}\right)nh_j^2 \leq \frac{22}{15}\kappa nh_j^2. \quad (14)$$

The qualifying endpoint is within $(\sqrt{22/15} + 1)\sqrt{n\kappa}h_j$ of $(y_j)_i$. Its continuous interval length is at most $h_j + \theta(\sqrt{22/15} + 2)\sqrt{n\kappa}h_j$, by comparison with the previous center. Since $\theta\sqrt{\kappa} \leq 1/\sqrt{8}$ and $n, \kappa \geq 1$, their sum is below $5\sqrt{n\kappa}h_j$. Taking the hull preserves this radius. An integer interval with positive correction obeys the same bound. A unit interval may survive through its good endpoint while its other endpoint is bad; its physical length contributes the stated extra one. $\square$

Lemma 4.3 (Uniform state bound). *Under the hypotheses of Lemma 4.1, every stage has at most*

$$K(\theta, n) = 100\theta^{-1}\lceil \log_2(n+2) \rceil \quad (15)$$

nodes per coordinate, independent of its stage number and accuracy.

Proof. Stage zero has at most three nodes per coordinate since $h_0 = s$. For continuous grids, the successive unclipped distances are $t_k = h((1 + \theta)^k - 1)/\theta$. A side of radius R therefore needs at most $1 + \lceil \log(1 + \theta R/h)/\log(1 + \theta) \rceil$ steps. For integer grids put $H = \max\{h, 1\}$. Every nonfinal integer step is at least $(H + \theta t)/3$: if $h \geq 1$, use $\lfloor u \rfloor \geq u/2$; if $h < 1$ split at $\theta t = 2$. Thus $t + H/\theta$ grows by a factor at least $1 + \theta/3$. The same count holds with H and $\log(1 + \theta/3)$.

At the next stage $h = h_j/2$. Lemma 4.2 bounds $\theta R/h$ in the continuous case and $\theta R/H$ in the integer case by quantities below $1 + 8\sqrt{n}$. Since $\log(1 + \theta/3) \geq \theta/4$ for $\theta \leq 1/4$, summing both side counts and the center is bounded by (15). The constant 100 includes ceilings and is not an optimized practical recommendation. $\square$

4.1 A capped schedule needs no supplied growth constant

For $\varepsilon > 0$, choose the smallest nonnegative integer J with

$$\frac{7Lns^2}{8} 4^{-J} \leq \varepsilon. \quad (16)$$

This uses rational comparisons, not logarithm oracles. Try $\theta = 2^{-\mu}$ for $\mu = 2, 3, \dots$, restarting from the original box and its lower endpoint vector as center each time. Run stages zero through J . While generating a grid, abort the trial before any coordinate exceeds (15), before allocating that trial's bag tables. Stop only when the verified global gap $U - b$ is at most ε .

The first trial with $\theta^2 \leq 1/(8\kappa)$ has $2^\mu \leq 6\sqrt{\kappa}$, passes the cap, and succeeds by Lemma 4.1. Earlier trials remain sound even when their guessed resolution is unsuitable. The caps bound their work by a geometric sum of state spaces. Without caps an earlier unsuccessful trial could pay an accuracy-dependent power before the successful one was reached.

The decisive absorption is

$$\lceil \log_2(n+2) \rceil^p \leq (C_0 p)^p (n+2) \quad (17)$$

for an absolute C_0 . It follows from $\sup_{t \geq 0} t^p e^{-t} = (p/e)^p$, adjusting the logarithm base and ceiling. A power of the accuracy bits could not be absorbed this way; the state bound has removed precisely that dependence.

Theorem 4.4 (Certified quadratic approximation). *For an explicit rational mixed-integer box quadratic with supplied decomposition and (3), the capped algorithm returns a rational feasible point and an independently checkable gap at most 2^{-q} in*

$$f(p, \kappa)(I + q + 1)^C \quad (18)$$

bit operations, where C is absolute. Neither g nor κ is an input. Certificate validity needs neither growth nor uniqueness.

Proof. The preceding lemmas prove correctness and bound state counts and stages. It remains to bound arithmetic. After fixed-coordinate substitution, choose a common positive denominator D for coefficients, endpoints, and L . Its bit length is polynomial in I . For a fixed trial let K be its cap, allowing one extra candidate for abort detection. At stage j , the unclipped continuous displacement is

$$h_j \frac{(2^\mu + 1)^k - 2^{\mu k}}{2^{\mu(k-1)}}.$$

All grid coordinates have denominators dividing $A_j = D2^{j+\mu K}$. Indeed inherited centers and endpoints have the previous common denominator, which divides this one; new displacements have the displayed denominator; clipping selects an inherited endpoint; integer floors stay integral. Denominators do not multiply over stages.

All quadratic values, corrections, and message entries have denominators dividing $8DA_j^2$. Their numerators have bit length polynomial in $I + j + \mu K$, since coordinates stay in the original box and each message is a sum of a subset of factor and correction values at some assignment. Factor evaluations, message arithmetic, comparisons, backtracking, certificate storage and checking have the same polynomial bit bound. The stage budget is $O(I + q + 1)$. Combine (8), (15), (17), and $2^\mu = O(\sqrt{\kappa})$, summing the earlier capped trials. All factors depending on p, κ belong to f , and the remaining polynomial exponent is absolute. $\square$

Approximation also terminates without growth. For a sufficiently fine trial $\theta \leq h_J/s$, each corrected interval at stage J is at most $2h_J$, so the gap is at most $Ln h_J^2/2 \leq 4\epsilon/7$. Before that stage continuous grids have at most $s/h_j + 3$ nodes, and integer grids at most $2s/h_j + 3$ when $h_j \geq 1$ or $s + 2$ otherwise. These fit the cap for sufficiently small θ . This observation gives termination, but not the growth-based rate.

5 Exact output is a separate theorem

5.1 Rational quadratics

Write the quadratic, after substituting fixed coordinates, as

$$F(x) = (x^\top H x + h^\top x + k)/D,$$

where H is symmetric integral, h, k are integral, and all endpoints have denominators dividing the positive integer D . Include factors of two from off-diagonal normalization when selecting D . Put

$$C_H = \max\{1, \max_{ij} |H_{ij}|\}, \quad \Delta = (2nC_H)^n, \quad R = D\Delta, \quad V = DR^2. \quad (19)$$

These numbers have polynomial bit length; they are not enumerated.

Lemma 5.1 (Rational height). *The problem has a global optimizer with a common coordinate denominator at most R , and its optimum value has reduced denominator at most V . Every isolated global optimizer satisfies the coordinate bound.*

Proof. Fix the integer assignment of a global optimizer. Among optimizers in that slice select one whose smallest containing continuous-box face has minimum dimension. Write S for its free continuous coordinates and A for integer and bound-active coordinates; $x_A = t_A/D$ with integral t_A . The free Hessian is positive semidefinite by local optimality. If singular, a free null direction has both zero first-order change and zero quadratic change. Move along it until a free coordinate hits a box bound. The objective stays optimal and the face dimension decreases, a contradiction. Thus $2H_{SS}$ is positive definite, and stationarity is

$$(2H_{SS})(Dx_S) = -Dh_S - 2H_{SA}t_A.$$

Cramer's rule gives common denominator $D \det(2H_{SS})$, bounded by $D|S|!(2C_H)^{|S|} \leq R$; an empty determinant is one. Evaluating the integer-coefficient quadratic at a common-denominator point gives reduced value denominator at most $DR^2 = V$. For an isolated optimizer a singular free Hessian would itself yield a short segment of optimizers, so the same argument applies directly. $\square$

Lemma 5.2 (Uniqueness supplies qualitative quadratic growth). *A quadratic with a unique optimizer on a compact mixed box has some $g > 0$ satisfying (3).*

Proof. Otherwise a sequence $x_j \neq x^*$ has ratio $(F(x_j) - F(x^*))/\|x_j - x^*\|^2 \rightarrow 0$. Compactness and uniqueness force $x_j \rightarrow x^*$, and hence their integer coordinates eventually equal those of x^* . Write $t_j = \|x_j - x^*\|$ and take a subsequence with $u_j = (x_j - x^*)/t_j \rightarrow u$, $\|u\| = 1$. The exact quadratic expansion is

$$\frac{F(x_j) - F(x^*)}{t_j^2} = \frac{\nabla F(x^*)^\top u_j}{t_j} + \frac{1}{2} u_j^\top \nabla^2 F u_j.$$

First-order optimality in the fixed integer slice makes the first term nonnegative. Its boundedness implies $\nabla F(x^*)^\top u = 0$, and the limiting quadratic term is nonpositive. The limit direction is in the

continuous-box tangent cone with zero integer components, so $x^* + \tau u$ is feasible for all sufficiently small positive τ . Optimality forces the same quadratic term nonnegative. It is zero, and the entire short ray is optimal, contradicting uniqueness. $\square$

This qualitative lemma does not bound conditioning usefully. A small input can encode an exponentially weak growth margin.

Theorem 5.3 (Exact rational quadratic output). *Under the hypotheses of Theorem 4.4, the unique optimizer and exact optimum value can be found and certified in $f_1(p, \kappa)(I + 1)^{C_1}$ bit operations for absolute C_1 , without receiving g . Every unique rational mixed-box QP has such a positive g , although its conditioning may be poor.*

Proof. Distinct rationals with denominators at most K differ by at least $1/K^2$. An interval $[b, U]$ of width at most $1/(4V^2)$ therefore isolates the unique possible optimum value of height at most V . Continued fractions reconstruct this rational in polynomial bit time. Around each coordinate of the feasible incumbent use a reconstruction window of radius $1/(4R^2)$. Its length is below $1/R^2$, so there is at most one rational of denominator at most R inside it. If all coordinates reconstruct, check the original bounds, integrality, and exact equality of the candidate's objective to the isolated value. Accept only after those checks. Acceptance is globally valid even before the incumbent is known to be close.

Call the approximation algorithm with $\varepsilon = 2^{-q}$ for $q = 1, 2, 4, 8, \dots$. When

$$\varepsilon \leq \min\{1/(4V^2), g/(32R^4)\},$$

growth places the incumbent within $1/(\sqrt{32}R^2) < 1/(4R^2)$ of x^* . Every coordinate reconstructs correctly and the check succeeds. The required q is $\text{poly}(I) + O(\log \kappa)$, using $g \geq L/\kappa$; doubling overshoots by at most two. The approximate bit bound and polynomial reconstruction cost prove the claim. The height constants come from the original problem, not from refined hull endpoints. $\square$

Only the optimal value is guaranteed to have denominator at most V . Arbitrary continuous feasible values do not occupy a fixed value lattice. The reconstruction and equality checks cannot be replaced by rounding an arbitrary small objective gap to zero.

Proposition 5.4 (Exact acceptance using the candidate's denominator). *Suppose $b \leq f^*$ is a certified original-domain lower bound and x is feasible. Write $F(x) = a/W$ in lowest terms, with $W > 0$. If*

$$F(x) - b < \frac{1}{VW},$$

then x is globally optimal. This implication requires neither uniqueness nor a growth constant.

Proof. The optimum is rational with reduced denominator at most V by Lemma 5.1. If $F(x) > f^*$, the difference of the two rationals is at least $1/(VW)$. Since $b \leq f^*$, that contradicts the displayed strict inequality. $\square$

This rule uses the denominator of the actual candidate; it does not assume that a feasible value has denominator at most V . It allows the implementation to verify candidates from reconstruction or exact stationarity recovery independently of how they were generated. A large candidate denominator can demand a very small certified gap, and the rule alone supplies no running-time bound for finding an accepted point.

Theorem 5.5 (Finite exact recovery without uniqueness). *Every nonempty bounded rational mixed-box QP admits finite exact output by the default geometric conditioning schedule, endpoint snapping and rational stationary-face recovery. The implementation succeeds with sufficiently large finite round, stage, table, LP-pivot and time limits. No growth constant or optimizer is supplied. This statement gives no general polynomial or fixed-parameter running-time bound.*

Proof. The height bound R can also bound vertices of every stationary polytope formed by fixing original integer labels and bounds and imposing a subset of free-gradient equations. In scaled coordinates $u = Dx$, a vertex matrix consists of integer stationary rows and unit bound rows. Expansion along the latter leaves a possibly nonprincipal Hessian minor. The same determinant estimate used in (19) bounds such a minor too. Thus each vertex has a common coordinate denominator at most R , including its original-bound slacks. The implementation’s sharper row-product bound has the same nonprincipal-minor property.

If no coordinates remain, direct evaluation is exact. Otherwise $n \geq 1$. Let S be the entire optimal set, $\tau = 1/(4nR)$, and suppose a feasible point y is within $\tau/2$ of a nearest optimizer s . Their integer labels agree. Snap each continuous coordinate within τ of an original endpoint to that endpoint. Original endpoint separations are at least $1/D > 2\tau$, so the rule is unambiguous and snaps every coordinate active at s correctly. Let J_0 be the interior continuous coordinates at s , and $J \subseteq J_0$ those left free after snapping.

Let P retain the original box, fix the labels and active coordinates of s , and impose $\nabla_{J_0} F = 0$. This nonempty bounded polytope has constant objective $F(s)$: differences of its points are supported on J_0 , and subtracting their gradient equations annihilates the free Hessian action. Let $q(x)$ sum the nonnegative slacks of the additional endpoints selected in $J_0 \setminus J$. Then

$$q(s) \leq 3n\tau/2 = 3/(8R) < 1/R.$$

A positive minimum of q at a vertex of P would be at least $1/R$, by the common denominator bound. Therefore some $s' \in P$ meets all selected endpoints simultaneously. The implemented selected-face stationarity system is feasible. Every feasible solution x of that system has the same value as s' : for $d = x - s'$, both free gradients vanish, d is supported on J , and $H_{JJ}d_J = 0$. Exact quadratic expansion gives $F(x) = F(s') = f^*$, without a nonsingularity assumption.

The unconditional approximation argument after Theorem 4.4 reaches every positive accuracy with sufficiently large finite caps. Compactness implies that feasible objective values tending to f^* approach S in distance. Thus sufficiently accurate incumbents satisfy the preceding recovery premise. Exact LP finds a recovery point even for singular faces, and a gap below $1/V^2$ makes Proposition 5.4 accept it. Each successful finite prefix uses finite resources. The proof supplies existence of such caps, not a useful bound on them for arbitrary optimal sets. $\square$

This specializes the earlier stationary-polytope recovery mechanism to the implemented denominator bound. It returns one rational optimizer and value. A compact representation of every optimizer, or an efficient width-and-set-growth bound for discovering an unknown optimal set, is a separate question.

5.2 Explicit polynomial factors

Suppose the factors are explicit rational monomial lists of fixed degree $d \geq 2$. A valid rational curvature certificate can be computed by differentiating twice in each coordinate, evaluating the resulting monomials by interval arithmetic on the full continuous hull, and summing their upper endpoints. A sharper checked bound may replace this one. Its certificate must have polynomial

verification cost; include the accepted L and any such certificate in I . The quantitative parameter uses the bound actually accepted.

Corollary 5.6 (Polynomial approximation and native-integer exactness). *At fixed numerical degree d , under (3), the pruned-grid algorithm returns a certified 2^{-q} approximation in $f_d(p, \kappa)(I + q + 1)^C$ bit operations with exponent independent of p, κ . If every coordinate is a native integer, it also returns an exact optimizer in $f_d(p, \kappa)(I + 1)^C$ bit operations.*

Proof. The interpolation, filtering, and growth proofs use coordinate semiconcavity, not a quadratic expansion. For arithmetic, polynomial values at stage j have denominators dividing DA_j^d , and values including corrections have a common denominator $8DA_j^{\max\{d, 2\}}$. Explicit fixed-degree evaluation is polynomial in input and coordinate bit length, which proves the approximation statement.

For native integers choose a common denominator Q_0 of the original objective coefficients. Every feasible value lies in $Q_0^{-1}\mathbb{Z}$. A certified gap less than $1/Q_0$ at a feasible incumbent excludes every strictly better integer value. It suffices to take $q = 1 + \lceil \log_2 Q_0 \rceil = O(I)$. The checker verifies feasibility, the original objective value, coefficient denominators, and the complete pruning certificate. Growth is unnecessary for this stopping rule’s validity. $\square$

The result does not automatically give exact continuous polynomial coordinates. For example $x^3 - 6x$ on $[1, 2]$ has optimizer $\sqrt{2}$ and value $-4\sqrt{2}$. Nor does uniqueness imply quadratic growth for polynomials: x^4 at zero is a counterexample. Arbitrarily large binary-encoded exponents or general circuits also need a separate evaluation-size analysis. A rational lattice rescaling can alter the conditioning parameter, so the native-integer theorem is not silently a metric-preserving lattice theorem.

6 Extensions and the remaining structural questions

6.1 Negative curvature: a useful metric does not yet give sparse states

For a continuous-box quadratic with Hessian H , let $\nu = \max\{0, -\lambda_{\min}(H)\}$. A stronger desired parameter would replace L/g by $\max\{1, \nu/g\}$, allowing large positive curvature to affect encoding length only. The target is continuous: $\nu = 0$ in a mixed-integer problem still permits convex integer optimization.

Two existing identities explain both the promise and the missing step. For $d = x - x^*$, first-order optimality on the convex box gives $F(x) - f^* \geq d^\top H d / 2$. Combining it with growth, with weights $g/(g + \nu)$ and $\nu/(g + \nu)$, yields

$$F(x) - f^* \geq \frac{g}{2(g + \nu)} d^\top (H + 2\nu I) d. \quad (20)$$

Thus the exact positive energy has the desired conditioning. A dense change of coordinates to make this energy Euclidean generally destroys the box factorization. Independent rounding need not preserve its cancellations. The inequality itself does not bound separator states.

Likewise, for $\lambda > \nu$ define the convex-recourse envelope

$$E_\lambda(c) = \min_{x \in X} \{F(x) + \lambda \|x - c\|^2 / 2\}.$$

Subtracting $\lambda \|c\|^2 / 2$ makes it a pointwise infimum of affine functions of c , hence concave. Its upper coordinate curvature is at most λ . Minimizing the lower bound $g \|x - x^*\|^2 + \lambda \|x - c\|^2 / 2$ over all $x \in \mathbb{R}^n$ gives

$$E_\lambda(c) - f^* \geq \frac{g\lambda}{2g + \lambda} \|c - x^*\|^2.$$

The ratio becomes $2 + \lambda/g$. Each rational evaluation is a strongly convex quadratic program, but the envelope can be dense even when H is tridiagonal. Efficient evaluation is not bounded-width factorization.

A conditional-moment obstruction. Keeping exact positive energy is useful but not sufficient by itself. For lifted moments with mean $m \in X$ and covariance $\Sigma = Y - mm^\top \succeq 0$, the relaxed objective satisfies

$$\frac{1}{2}\langle H, Y \rangle + b^\top m + c = F(m) + \frac{1}{2}\langle H, \Sigma \rangle \geq F(m) - \frac{\nu}{2} \text{tr } \Sigma.$$

If these moments are conditioned on a single global cell of widths w_i , interval moment bounds give $\Sigma_{ii} \leq w_i^2/4$ and loss at most $\nu \sum_i w_i^2/8$. A mixture indexed by complete global cell choices inherits the same weighted bound. Matching finitely many separator moments in individual bag cells does not ensure this representation.

An explicit witness uses $(s, u_1, u_2, v) \in [0, 1]^4$, rational $0 < h \leq 1/2$, and

$$F_h = 8(s/h - (u_1 + u_2)/2)^2 + 8(s/h - 1/4 - v/2)^2 \\ + u_1(1 - u_1) + u_2(1 - u_2) + v(1 - v) + (u_1 + u_2 + v)/16.$$

The bags are $\{s, u_1, u_2\}$ and $\{s, v\}$. With $\delta = s/h - 1/8 - (u_1 + u_2 + v)/4$, expansion gives

$$F_h - \frac{1}{4} = 16\delta^2 + 2[(1 - v)u_1u_2 + v(1 - u_1)(1 - u_2)] + (u_1 + u_2 + v)/16.$$

This proves unique optimum $(h/8, 0, 0, 0)$, value $1/4$, and growth $g = 1/22$: the squared distance is at most $2\delta^2 + 11\|(u_1, u_2, v)\|^2/8$. The Hessian is PSD squares plus diagonal negative curvature -2 on controls; direction $(0, 1, -1, 0)$ attains eigenvalue -2 , so $\nu/g = 44$.

For $t = s/h$, put left-bag masses $1/8, 3/8, 3/8, 1/8$, respectively, at

$$(t, u_1, u_2) = (0, 0, 0), (1/2, 1, 0), (1/2, 0, 1), (1, 1, 1).$$

Put equal right-bag masses at $(t, v) = (1/4, 0), (3/4, 1)$. Both separator laws have moments $\mathbb{E}t = 1/2$, $\mathbb{E}t^2 = 5/16$, $\mathbb{E}t^3 = 7/32$. All local squares and concave penalties vanish, giving relaxed value $3/32$ and gap at least $5/32$. On mesh $2h$, every separator atom lies in its first cell and controls occupy endpoint cells; all occupied widths tend to zero. Even a global PSD completion exists: with common mean $(h/2, 1/2, 1/2, 1/2)$ take

$$\Sigma = aa^\top/16 + 3bb^\top/16, \quad a = (h, 1, 1, 2)^\top, \quad b = (0, 1, -1, 0)^\top.$$

Its bag restrictions match the measures. Thus a width-squared error bound depending only on $p, \nu/g$ fails for this particular local-moment interface. It is not a lower bound against adaptive filtering or all moment hierarchies; matching the fourth separator moment already excludes this witness. The companion supplies the complete construction and checks, including global McCormick inequalities.

6.2 A complete reduction for certified affine convex recourse

There is a useful class on which stiff positive curvature can be removed exactly. Retain z in a rational product box and let private, mutually disjoint blocks $y_t \in [\ell_t, u_t]$ attach only to coordinates $E_t z$. For one block, suppress t and write

$$f(y, z) = \frac{1}{2}y^\top Cy + y^\top Dz + c^\top y + \frac{1}{2}z^\top Az + d^\top z + e, \quad C \succeq 0.$$

Supply affine maps $\bar{y}(z) = a + Bz$, $\lambda(z)$, and $\mu(z)$ that satisfy, throughout the attachment box,

$$\begin{aligned}\ell &\leq \bar{y}(z) \leq u, & \lambda(z), \mu(z) &\geq 0, \\ C\bar{y}(z) + Dz + c &= \lambda(z) - \mu(z), \\ \lambda_i(z)(\bar{y}_i(z) - \ell_i) &= 0, & \mu_i(z)(u_i - \bar{y}_i(z)) &= 0.\end{aligned}$$

Affine box ranges, coefficient identities, and rational PSD elimination check these conditions in polynomial bit time, including singular C . Put $q_t(z) = f_t(\bar{y}_t(z), z)$. Quadratic expansion and complementarity give the exact nonnegative identity

$$f(y, z) - q_t(z) = \frac{1}{2}(y - \bar{y})^\top C(y - \bar{y}) + \lambda(z)^\top(y - \ell) + \mu(z)^\top(u - y) \geq 0. \quad (21)$$

Equality holds at $\bar{y}(z)$. Thus minimizing the original objective is exactly minimizing $q(z) = q_0(z) + \sum_t q_t(E_t z)$, whose block Hessian is $A + B^\top C B + B^\top D + D^\top B$.

If the original problem has point growth g , evaluating that inequality on the affine response yields

$$q(z) - f^* \geq g(z - z^*)^\top \left(I + \sum_t E_t^\top B_t^\top B_t E_t \right) (z - z^*). \quad (22)$$

In particular Euclidean growth g transfers. If every attachment fits in a supplied residual bag of size at most p , reduced factors retain that width. If all reduced Hessian diagonals are nonpositive, endpoint DP is exact without growth. Otherwise set $L_{\text{red}} = \max_i (\nabla^2 q)_{ii} > 0$; Theorem 5.3 applies with this reduced upper coordinate curvature and $\kappa_{\text{red}} = \max\{1, L_{\text{red}}/g\}$; the affine map lifts its exact output. Identity (21) transfers every lower bound and certificate to the original problem independently of growth. When $L_{\text{red}} \leq C_0 \beta$, for a checked rational $\nu \leq \beta < 2\nu$ and fixed C_0 , this is a scoped algorithm parameterized by p and $\max\{1, \nu/g\}$. The condition on reduced positive curvature remains an additional hypothesis.

Fixed-active-set affine recourse is standard multiparametric quadratic programming [2]. The contribution here is its checked composition with the conditioned sparse algorithm and metric growth transfer. The relevant width is the residual width after elimination; original width alone does not control fill-in. The companion note also gives a ladder with arbitrarily many negative eigenvalues, fixed reduced conditioning, and arbitrarily stiff penalties $M \sum_i (y_i - u_i)^2$. Its algebraic certificate demonstrates the parameter separation without claiming a difficult benchmark. General changing active faces and their curvature cancellations require further structure; a general negative-curvature-only bound is still open.

The global affine map need not be supplied for a fixed private-block partition. Choose the strict interior midpoint z_0 of the attachment box and solve one exact central convex QP, obtaining $\hat{y}$. All central minimizers have the same gradient $h = C\hat{y} + Dz_0 + c$: PSD expansion between two minimizers forces their difference into $\ker C$. Coordinates with $h_i > 0$ must be at their lower bounds, those with $h_i < 0$ at upper bounds. Any globally affine feasible selector fixes these coordinates at those bounds, because an affine bounded function attaining a bound at an interior parameter is constant. For $h_i = 0$, the selector's gradient must vanish identically. If its coordinate is not identically a bound, it is interior at interior parameters, and stationarity applies; if it is constant at a bound, its affine gradient has one sign and vanishes at the midpoint, so is zero everywhere. Thus existence is equivalent to a polynomial-size linear feasibility system for affine selector coefficients: these forced bounds, zero gradient identities, full-box primal bounds and the remaining gradient signs. Absolute-value auxiliary variables encode affine box ranges. One exact central QP and one LP therefore decide existence and construct the rational certificate, including singular PSD blocks. A failed test rejects this affine-selector class, not the underlying optimization problem. The companion gives the full recognition proof; choosing a useful private-block partition remains outside it.

Changing active sets without explicit value-function pieces. A second exact composition keeps private convex blocks as value factors. Let Y_t be fixed nonempty bounded rational polytopes, independent of the retained variables, and define

$$\psi_t(v) = \min_{y \in Y_t} \{ \frac{1}{2} y^\top C_t y + y^\top D_t v + c_t^\top y \}, \quad C_t \succeq 0.$$

For every fixed y the parameter dependence is affine, so ψ_t is concave. In $V(z) = q_0(z) + \sum_t \psi_t(E_t z)$, the upper coordinate curvature is bounded by the direct quadratic diagonals of q_0 . Every rational value query is an exact convex QP with primal and rational KKT certificates [10]. The attachment scopes fit a supplied residual decomposition; the number of retained variables need not be small. Original full-vector growth implies retained-coordinate growth by evaluating it at conditional minimizers.

The corrected-grid proof now applies with local convex-QP evaluations instead of polynomial factor evaluation. Different active sets can give different denominators, but each message is a selected sum of at most the input number of polynomial-bit oracle values. Its denominator bit length is bounded by their summed lengths, so the same parameterized approximation bound holds. Exact recovery uses height bounds for the *original* rational QP on a bounded polytope, not for the nonsmooth value function. Reconstruct the retained vector, lift through the exact oracles, and accept only at the isolated original optimum value. The companion proof establishes the full bit and height argument. This handles changing active faces but need not cancel large direct positive curvature: for $M(y - z)^2$ the direct diagonal is $2M$. Affine condensation can first remove blocks with a global response certificate, leaving other blocks as value factors.

Local pieces can preserve curvature cancellation. The affine and changing-active-set cases admit a stronger common interface. For a private convex block write

$$f(y, v) = \frac{1}{2} y^\top C y + y^\top D v + c^\top y + \frac{1}{2} v^\top A v + d^\top v + e, \quad C \succeq 0.$$

Suppose a rational binary space partition covers its attachment box. Every closed leaf carries an affine feasible response $\bar{y} = a + Bv$ and affine nonnegative bound multipliers satisfying stationarity and complementarity throughout that leaf. A partition tree proves coverage; rational LP dual witnesses prove the affine inequalities. The identity

$$f(y, v) - f(\bar{y}(v), v) = \frac{1}{2} (y - \bar{y})^\top C (y - \bar{y}) + \lambda(v)^\top (y - \ell) + \mu(v)^\top (u - y) \geq 0$$

then verifies the reduced quadratic on that entire leaf. Distinct response maps may agree only in value at a shared boundary, which is sufficient. Let $H_r = A + B_r^\top C B_r + B_r^\top D + D^\top B_r$ be the reduced Hessian on leaf r and put $\ell_j = \max_r (H_r)_{jj}$.

Proposition 6.1 (Curvature across a certified partition). *The conditional value $\phi(v) = \min_y f(y, v)$ has upper coordinate curvature ℓ_j in coordinate j . Therefore local partitions for independent blocks give the valid reduced coordinate bound*

$$L_i = (H_0)_{ii} + \sum_{t: i \in E_t} \max_r (H_{tr})_{j(t,i), j(t,i)},$$

where H_0 is the retained quadratic's Hessian. No global overlay of the local partitions is needed.

Proof. Restrict a coordinate line. The finite partition makes ϕ continuous and piecewise quadratic, with second derivative at most ℓ_j within each piece. Moreover $\phi(v) - v^\top A v / 2 - d^\top v - e$ is an infimum

of affine functions of v , hence concave. Its derivative jumps on the line are downward. Adding or subtracting a smooth quadratic does not change those jumps. Consequently the derivative of $\phi - \ell_j v_j^2/2$ is nonincreasing within and between pieces. The same argument holds on a partition boundary, where overlapping pieces agree as restricted polynomials. Summing the coordinate bounds proves the displayed formula. $\square$

Let S include the encoding lengths of all partition and KKT witnesses. With unique full-vector growth g and residual maximum bag size p , the corrected-grid algorithm now has the previous bounds $f(p, \max\{1, L/g\}) \text{poly}(S + q)$ for approximation and $f_1(p, \max\{1, L/g\}) \text{poly}(S)$ for exact rational output, where $L = \max_i L_i > 0$. Indeed, a factor query traverses one local tree, original growth implies retained growth, and exact output uses the original QP height. If every $L_i \leq 0$, endpoint DP is exact without growth. Certificate size is part of the input to this statement; constructing a small partition from an arbitrary QP is not guaranteed.

For a concrete separation, take $M \geq 1$, $0 \leq z \leq 3/4$ and $0 \leq y_1, y_2 \leq 1$, with

$$f_M = M(y_1 + y_2 - z)^2 + (y_1 - 2z + 1/2)^2.$$

The three parameter intervals have reduced values $(1/2 - 2z)^2$ on $[0, 1/4]$, zero on $[1/4, 1/2]$, and $M(z - 1/2)^2/(M + 1)$ on $[1/2, 3/4]$. Their curvature maximum is 8, whereas the direct retained diagonal is $2M + 8$. The unique conditional response changes slope, so no globally affine response represents it. The companion note extends this to a ladder with arbitrarily many negative eigenvalues, uniformly bounded negative curvature and full-vector growth, and three pieces per block. It supplies that family's complete parameter proof and the finite construction by active patterns. The implemented constructor and independent checker cover scalar attachments; the general certified-partition theorem does not assert an implemented automatic multidimensional partition algorithm. Parametric critical regions themselves are classical [2].

6.3 Why a global grid correction cannot simply be made local

For $m = 32$ and $\epsilon = 1/16$, on $[0, 1]^{m+1}$ consider

$$F(x, y) = x^2 - \frac{x}{2} \sum_{i=1}^m y_i + \sum_{i=1}^m y_i^2 + \left(\frac{m}{16} - 1 - \epsilon \right) x.$$

This is a star with bags $\{x, y_i\}$ and coordinate curvature $L = 2$. Exact leaf recourse has $y_i = x/4$ and value $V(x) = -x^2 + (1 - \epsilon)x$, so its unique optimizer has $x = 1$, $y_i = 1/4$, and value $-1/16$. On the grid $\{0, 1/2, 1\}$ the conditional grid values at the three center labels are 0, $23/32$, and $31/16$. The grid incumbent is zero. For the bag cell $x \in [1/2, 1]$, $y_i \in [0, 1/2]$, which contains the optimizer's bag projection, the smallest corner grid min-marginal is $23/32$. A proposed bag correction $|B|Lh^2/8 = 1/8$ would reject the cell. The missing outside error is

$$V_h(x) - V(x) = m \text{dist}(x/4, \{0, 1/2, 1\})^2,$$

which is not constant in the separator and does not disappear under message normalization. This refutes that local substitution, not every possible conditional-recourse method.

6.4 The precise conditional-recourse interface

For a bag B , define the true original-domain value $W_B(v) = \min_{x_{-B}} F(v, x_{-B})$. When the outside feasible set is independent of v , pointwise infima preserve coordinate semiconcavity: after subtracting

$Lv_i^2/2$, each member is concave, and the infimum of concave functions is concave. For a bag cell of side at most h , rounding only its bag coordinates gives lower bound

$$\min_{v \text{ a corner}} W_B(v) - e_B, \quad e_B = |B|Lh^2/8.$$

No outside variables are rounded.

Suppose an oracle gives $\ell_B(v) \leq W_B(v) \leq u_B(v)$, a feasible completion attaining $u_B(v)$, and $u_B(v) - \ell_B(v) \leq \eta \leq ae$, where $e = pLh^2/8$ is a common budget. First update the incumbent from all current-level completions and then retain cells whose corner lower bound minus e is at most U . Interpolation at an optimizer gives $U \leq f^* + e + \eta$. Every retained cell has a corner with

$$W_B(v) \leq f^* + 2(e + \eta).$$

This is the sufficient local-error interface. Obtaining these bounds requires certified optimization over the original outside domain; ordinary grid messages do not satisfy the interface automatically.

6.5 An exact cut oracle for a large nonconvex residual

The conditional interface has a concrete realization. For

$$F(x) = c + \sum_i (a_i x_i^2 + b_i x_i) + \sum_{i < j} c_{ij} x_i x_j,$$

supply a core C of k continuous variables and residual R such that $a_i \leq 0$ for $i \in R$ and signs $s_i \in \{-1, 1\}$ satisfy $c_{ij} s_i s_j \leq 0$ for residual edges. A graph traversal checks the sign consistency for a supplied core. There are no restrictions on edges touching the core, and the residual graph may be dense.

Fix a rational core assignment. Successive coordinatewise concave endpoint choices show that a residual optimum exists at endpoints. For integer residual coordinates round their bounds inward first; the same endpoint argument applies without enumerating their integer labels. Choose signed endpoint representations $x_i = \alpha_i + d_i y_i$ with $y_i \in \{0, 1\}$ and $d_i = s_i(u_i - \ell_i)$. The residual energy is

$$K + \sum_i h_i y_i + \sum_{i < j} q_{ij} y_i y_j, \quad q_{ij} = c_{ij} d_i d_j \leq 0.$$

With $r_{ij} = -q_{ij}/2$ and $p_i = h_i + \frac{1}{2} \sum_{j \neq i} q_{ij}$, this equals $K + \sum_i p_i y_i + \sum_{i < j} r_{ij} |y_i - y_j|$. Represent $y_i = 1$ on the source side of a cut. Arcs $i \rightarrow t$ of capacity $\max\{p_i, 0\}$, $s \rightarrow i$ of capacity $\max\{-p_i, 0\}$, and opposite pair arcs of capacity r_{ij} give

$$F = K + \sum_i \min\{0, p_i\} + \text{cut}(y). \tag{23}$$

A feasible flow of the same value as the returned cut proves conditional optimality by weak duality. Rational max-flow is polynomial in query encoding length, and the identity, flow and cut need only linear-size checks after graph construction. The class is stable under residual subboxes and added linear terms.

The cut reduction is classical [9]; its role here is a certified outside oracle that removes the entire residual dimension from discretization. Separate concavity is essential to this route. General continuous quadratic submodularity is not being treated as an exact cut oracle. In particular the current semidefinite-representation paper proves exactness in dimensions at most three and exhibits a four-dimensional gap [4].

The companion mixed-submodular result broadens this residual class when, after sign reversal, the nonpositive-diagonal variables can be sent to endpoints and the remaining continuous principal block is PSD. Minimizing that convex block gives a submodular set function of the endpoint labels: apply the lattice inequality to minimizing completions and their coordinatewise minimum and maximum. Exact submodular minimization and convex-QP queries then give polynomial-time recourse. Its polynomial certificate uses a convex combination of at most $|R| + 1$ greedy base vectors, certified convex-QP chain values, and matching primal value. The greedy-base certificate is also classical [8]; the added contract is exact convex-QP witnesses and box-stable composition. The full constructive certificate proof remains in the companion note. The completed reference backend implements rational cutting planes for the Lovász extension, conditional convex-QP queries and the resulting certificate. Its finite capped implementation is distinct from the polynomial-time submodular oracle used in the complexity theorem.

For core domain $[0, 1]^k$, query corner values of $V(v) = \min_{x_R} F(v, x_R)$ on nested dyadic core cells. With core upper coordinate curvature L , a side- h_j cell has error $e_j = kLh_j^2/8$. Update the incumbent using all queried completions, retain cells with $\min_{\text{corners}} V - e_j \leq U$, and generate only children of retained cells. This has a complete trace certificate and gap at most e_j , by the interface proved above.

There is also a precise smoothed count. Perturb core linear coefficients independently and uniformly on M equally spaced points of $[-\sigma, \sigma]$, with $M \geq \max\{2, 2^J\}$ chosen before sampling. For a fixed level- j tuple with value at most $f^* + 2e_j$, comparison with both coordinate neighbors confines an interior coordinate's noise to an interval of length at most $Lh_j + 4e_j/h_j = Lh_j(1 + k/2)$. The interval depends on the unperturbed conditional value and the fixed tuple, not on other core noises. Its probability is at most its length divided by 2σ , plus $1/M$. Summing the independent product over all full-grid tuples bounds their expected number by

$$H_k = \left[3 + \frac{(1 + k/2)L}{2\sigma} \right]^k.$$

Every retained cell touches such a tuple; a tuple touches at most 2^k cells, each has 2^k children and each child 2^k corners. Hence the expected number of conditional queries through level J is at most $2^k + 8^k J H_k$. Multiplying by polynomial query and certificate cost gives an absolute-exponent polynomial in residual size and precision, with the exponential dependence confined to k and L/σ . Correctness holds on every draw, and the optimized objective is the perturbed objective. This realizes the conditional interface on a supplied-core class; it does not solve the corresponding question on arbitrary sparse graphs.

Implemented exact quadratic output. This oracle also permits deterministic exact output with no growth assumption, although the cell count can be large. Let S_0 be the product of denominators of objective coefficients and original box endpoints, $T_0 = S_0^3$, and $H_0 \geq 1$ the largest absolute entry of the integer scaled core Hessian $T_0 \nabla_{CC}^2 F$. For any residual endpoint label, $T_0 F$ is an integer-coefficient quadratic in the unit-box core. A smallest-face optimizer has nonsingular free Hessian, as in Lemma 5.1. Its determinant is at most $k! H_0^k$, so its value denominator is at most

$$D_0 = T_0 (k! H_0^k)^2.$$

The same bound applies to the global value without enumerating all residual labels. Run core search to gap $1/(2D_0^2)$. Fix the incumbent's residual label and solve its core QP by enumerating its 3^k faces and nonsingular stationary systems, including vertices. The minimum-face argument makes this finite list complete. Its value $\hat{v}$ lies between f^* and U , and both $\hat{v}$ and f^* have denominators

at most D_0 . Rational separation forces equality. The checker only verifies the search trace, height bound, candidate feasibility, candidate value denominator at most D_0 , and that value's membership in the narrow interval; it need not repeat face enumeration. The implemented mode reports an unfinished enclosure if its level budget is insufficient. This stopping depth must not simply be inserted into the finite-noise count: sampled coefficients affect D_0 , and requiring $M \geq 2^{J(D_0(M))}$ could be circular. The separate exact smoothed theorem uses an existing base-selected closure and fallback construction, not that substitution.

6.6 Coupled constraints and nonunique optima

The product-domain hypothesis does mathematical work. Independent rounding may violate an equality even when the equality is sparse. Constraint extensions must supply a feasible rounding law, an exact reduction, or recourse with a verified error budget. A repair map must also account for distortion of objective error, conditioning, and integrality. Merely placing constraint scopes in small bags does not complete this argument.

For a nonunique optimal set S , set growth $F(x) - f^* \geq g \text{dist}(x, S)^2$ need not localize all retained coordinates near one center. The point-growth proof above therefore cannot be relabelled as a set-growth proof. Special structure or an explicit selection mechanism is needed. Exact polynomial boundary output likewise requires a certificate appropriate to active faces and algebraic coordinates; quadratic rational reconstruction is not such a certificate.

6.7 A complete extension for integral TU fibers

Let

$$X = \{(x, z) : \ell \leq x \leq u, z_i \in Z_i, Ax + Bz \leq b\},$$

where A is totally unimodular (TU), ℓ, u, B, b are integral, and each Z_i is an explicitly listed finite integer set. Constraint rows and objective factors fit supplied bags of maximum size p . Let n_c count nonfixed continuous coordinates, $d = \max_i |Z_i|$, and $W = \max_i (u_i - \ell_i)$. Assume a verified full directional bound $\nabla_{xx}^2 F \preceq LI$ on the continuous hull, uniformly over finite labels, and unique-point growth g on X .

At level j use aligned mesh $h = 2^{-j}$, keeping endpoints of retained continuous hulls and all retained finite labels. For fixed feasible (x, z) write its surrounding cell as $x = h(k + t)$. The fiber is

$$At \leq 2^j(b - Bz) - Ak, \quad 0 \leq t \leq 1.$$

Its right side is integral and adjoining box bounds preserves TU. Therefore every vertex is binary, and t is a convex combination of feasible binary corners. This is classical box integrality [5]. The resulting feasible random corner Y has $\mathbb{E}Y = x$ and $\mathbb{E}\|Y - x\|^2 \leq n_c h^2 / 4$. Taylor's upper bound gives

$$\mathbb{E}F(Y, z) \leq F(x, z) + E_j, \quad E_j = n_c L h^2 / 8. \quad (24)$$

Correlations in this rounding explain why a diagonal-curvature bound alone would not suffice.

Exact tree DP rejects infeasible bag assignments and computes the least feasible grid objective v_j and coordinate min-marginals M_i . Then $b_j = v_j - E_j$ is a lower bound. The current incumbent lies on each later nested grid, so after updating it $U = v_j$. Retain a continuous interval $[s, t]$ when $\min\{M_i(s), M_i(t)\} - E_j \leq U$, and a finite label t when $M_i(t) - E_j \leq U$. Conditional feasible rounding proves these exclusions sound and gives the same full-history certificate argument as Proposition 3.2. An empty initial grid certifies infeasibility by the same rounding lemma.

A retained endpoint or label has a full feasible grid witness with value at most $U + E_j \leq f^* + 2E_j$. Growth puts it within $r_j = \sqrt{n_c L / (4g)} h$ of the optimizer. Continuous retained hulls therefore have length at most $2r_j + 2h$, and the next grid has at most $5 + 2\sqrt{n_c L / g}$ nodes per continuous coordinate. Thus table work is

$$O(p(N + |\mathcal{A}| + \#\text{rows})[K_0^p + (J + 1)K^p]), \quad (25)$$

where $K_0 = \max\{d, W + 1\}$, $K = \max\{d, 5 + \lceil 2\sqrt{n_c \kappa} \rceil\}$, and the least J with $E_J \leq 2^{-q}$ is $O(I + q + 1)$. Dyadic coordinates and fixed-degree explicit evaluation have polynomial arithmetic overhead. No growth constant is used by this algorithm or its checker.

For rational quadratics, finite-label fixation and shrinking continuous hulls allow exact rational reconstruction. On a minimum-dimensional optimal continuous face, the Hessian is positive definite on the face tangent space by the null-direction argument of Lemma 5.1. The integral KKT saddle matrix has dimension at most $2n_c$ and bounded entries, so Cramer's rule gives a polynomial-bit common denominator bound. Reconstruct only after hulls isolate bounded-height coordinates and the value interval isolates a bounded-height value; then check all original constraints and the exact objective. The companion TU note supplies explicit determinants and thresholds. The level count becomes $\text{poly}(I) + O(\log \kappa)$ with the same state factors.

This is a complete accuracy-independent state theorem, but it is *not* FPT in (p, κ) : the factor $n_c^{p/2}$ remains. Large integral capacities incur the initial pseudopolynomial K_0^p cost. Network conservation with unit capacities and integral finite-label switch contributions is a concrete application; every conservation row still contributes its full scope to width. General nonlinear constraints, unaligned rational fibers, and compressed large finite-label sets are outside this statement.

Two reviewed refinements keep the original variables and bag scopes. First, with original equalities $Cx + Ez = a$, the curvature bound is needed only for directions in $\ker C$, because feasible rounded corners have the same labels and preserve those equalities. For a quadratic, a rational basis of $\ker C$ or its orthogonal projector verifies the restricted bound without substituting variables into objective factors. A penalty $\rho \|Cx + Ez - a\|^2$ then changes neither grid values nor rounding allowances. Coefficient-based exact height bounds can still grow with ρ , so this statement concerns per-level search and correction. Second, a supplied positive rational initial mesh δ is admissible when $\ell/\delta, u/\delta, (b - Bz)/\delta$ are integral for all listed labels. Check one reference label vector and each individual column-label increment, without enumerating their Cartesian product. The initial count becomes $W/\delta + 1$, while the filtered count is unchanged and $h_j = \delta 2^{-j}$. Large common units can therefore remove the initial capacity cost; arbitrary unaligned magnitudes still retain that cost.

Nonunique TU minima and unions of retained cells. Exact output can be extended to arbitrary nonunique TU quadratics, and a stronger state representation helps when each continuous coordinate has only finitely many optimal values. Let $S = \arg \min_X F$ and use the quadratic error bound $F(v) - f^* \geq g \text{dist}(v, S)^2$ for some $g > 0$. Such a constant exists on each compact quadratic/polytope slice [12]; finitely many other integer slices have positive value gaps. No useful numerical lower bound on g is asserted.

Keep the union of surviving mesh cells, including singleton nodes, instead of their interval hull. Every feasible point still lies in one mesh cell within the retained domain, so the same TU corner distribution and error E_j apply. In particular

$$b_j = v_j - E_j \leq f^* \leq U_j = v_j \leq f^* + E_j.$$

A passing endpoint has a feasible witness of value at most $f^* + 2E_j$, and hence lies within $a_j = \sqrt{n_c L / (4g)} h_j$ of some optimal coordinate value. Every point of a retained cell lies within

$a_j + h_j$ of such a value. If the projection S_i has at most r elements, the next mesh therefore has at most

$$r(5 + \lceil 2\sqrt{n_c \kappa} \rceil)$$

labels in coordinate i . This replaces the continuous part of K in (25). Different coordinates may have witnesses near different optimizers; the argument neither enumerates nor chooses optimal combinations. There can be exponentially many isolated optima. Positive-dimensional continuous optimal sets are excluded by the finite-projection state bound, but not by the exact-output statement that follows.

For exact output, the general-polytope recovery lemma in the companion work computes original-data rational thresholds $\delta, \tau > 0$ and an optimum denominator bound V , all of polynomial binary length. Given a feasible point within distance δ of S , fix its integer labels, impose original inequality rows with slack at most τ as equalities, and solve rational feasibility with stationarity along that face. Every feasible recovery output is optimal. The lemma’s proof uses stationary-polytope heights and simultaneous slack snapping; this is an imported recovery result, not a claim that arbitrary KKT points are global. Its stationarity multipliers are unrestricted.

Refine until $E_j < 1/(4V^2)$, isolate the bounded-height optimum value in $[b_j, U_j]$, and try the recovery LP. Accept a point only after checking all original constraints and equality to that isolated value. Early recovery can fail or produce a nonglobal stationary point, but neither can pass the final equality check. Eventually $E_j \leq g\delta^2$, so recovery succeeds, including for a continuum of optimizers. The required level is $\text{poly}(I) + O(\log \kappa)$; favorable grid-size bounds additionally require finite projections. This extends exact termination without a supplied growth constant. It does not resolve the stronger width-FPT problem for arbitrary unknown optimal sets.

6.8 Compact descriptions of nonunique optimal sets

Two restricted results avoid pretending that all retained coordinates converge to a single unknown optimum.

First suppose $F(x) = x^\top Qx + b^\top x + c$ on a mixed box and $Q_{ii} \leq 0$ for every coordinate. Endpoint DP gives root value f^* . Root the decomposition and let m_t denote its upward message. For assigned bag factor ϕ_t , define the nonnegative endpoint Bellman residual

$$r_t(v_{B_t}) = \phi_t(v_{B_t}) + \sum_{c \text{ child}} m_c(v_{S_c}) - m_t(v_{S_t}) \geq 0,$$

where the root message is f^* . On any endpoint assignment the messages telescope to $F(v) - f^* = \sum_t r_t(v_{B_t})$. Let $W_t(v; x)$ be the product of endpoint interpolation weights $(u_i - x_i)/(u_i - \ell_i)$ or $(x_i - \ell_i)/(u_i - \ell_i)$. Independent mean-preserving endpoint rounding preserves cross terms and adds variance $(x_i - \ell_i)(u_i - x_i)$ to x_i^2 . Taking expectations therefore gives the exact identity

$$F(x) - f^* = \sum_t \sum_{v \in \{\ell, u\}^{B_t}} r_t(v) W_t(v; x) - \sum_i Q_{ii} (x_i - \ell_i)(u_i - x_i). \quad (26)$$

All terms are nonnegative. Optimality is equivalent to mixed feasibility, vanishing of every positive-residual product, and endpoint status for every $Q_{ii} < 0$. This is a factored polynomial description of all optimizers with at most $N2^p + n$ equations. It is constructed and checked in $2^{O(p)} \text{poly}(I)$ bit work with no growth promise and no enumeration of interior integer labels or optimal components. Endpoint reduction and Bellman reparameterization are classical; the displayed identity records their full-set certificate.

Second, for a continuous-box quadratic $F(x) = \frac{1}{2}x^\top Hx + b^\top x + c$ and a feasible box KKT point s , define

$$d_i(s) = \begin{cases} 2\partial_i F(s)/(u_i - \ell_i), & s_i = \ell_i, \\ -2\partial_i F(s)/(u_i - \ell_i), & s_i = u_i, \\ 0, & \ell_i < s_i < u_i, \end{cases} \quad D = \text{diag}(d_i).$$

These entries are nonnegative. Direct quadratic expansion yields

$$F(x) - F(s) = \frac{1}{2}(x - s)^\top (H + D)(x - s) + \frac{1}{2} \sum_i d_i(x_i - \ell_i)(u_i - x_i). \quad (27)$$

If $H + D \succeq 0$, this certifies global optimality. The condition is the classical box Lagrangian sufficient condition of Li, Wu, and Quan [11], not a new certificate class. It gives the full optimal set by

$$\ell \leq x \leq u, \quad (H + D)(x - s) = 0, \quad d_i(x_i - \ell_i)(u_i - x_i) = 0 \quad \text{for all } i.$$

If the test passes at one optimum, it passes at every optimum with the same D : equality in (27) puts $t - s$ in the kernel and forces each positive- d_i coordinate to an endpoint; $\nabla F(t) = \nabla F(s) - D(t - s)$ then gives exactly the same entries.

The continuation combines this acceptance test with the earlier proximal grid candidate generator and exact selected-face recovery lemma. Doubling untrusted guesses K for L/g , running each generator for its prescribed finite budget, and accepting only a rational KKT/PSD certificate is sound under every guess. On instances with at least one certificate of this form, the first $K \geq \max\{1, L/g\}$ produces a globally optimal candidate, and the invariance just proved ensures acceptance whichever optimal component was recovered. The resulting exact discovery and full-set description take $f(p, \max\{1, L/g\}) \text{poly}(I)$ bit work under set growth toward S . Neither S nor g is supplied. The candidate generator and recovery proofs remain in the identified companion notes; the acceptance and full-set identity are proved here. Outside this certifiable class the procedure may remain inconclusive.

6.9 Polynomial boundary output with an explicit margin

There is a restricted way to remove the full-Hessian positivity premise from exact implicit polynomial output. Assume unique-point growth and strict complementarity at every active continuous original bound. Let $\gamma > 0$ be the smallest inward derivative there, and let a checked rational $M \geq 1$ bound all absolute continuous-Hessian row sums on the original hull. Put $B_\gamma = \lceil \log_2 \max\{2, M/\gamma\} \rceil$ for analysis only.

After retained integer labels have become singletons, work within their fixed slice. In a globally retained continuous box of maximum half-width r and midpoint c , the bound $\partial_i F(c) - Mr > 0$ certifies fixing a lower-bound coordinate; the corresponding negative upper bound certifies an upper face. Each substitution preserves the minimum on the retained box. An optional exact Bernstein coefficient bound also permits weak signs, preserving at least one optimizer. This is standard monotonicity propagation [1], now combined with sparse containment certificates.

At $r \leq \gamma/(4M)$ all active coordinates are found. The remaining free Hessian at the optimizer is at least $2gI$, since two-sided free directions and the growth inequality give that second-order bound. A rational third-derivative bound T lets an exact midpoint test $H_B(c) - (Tr + \tau)I \succ 0$ certify a strongly convex patch on the selected face. Dovetail capped grid trials with $\theta_\mu = 2^{-\mu}$, $K_\mu = 4^\mu$, and $\tau_\mu = L/(4K_\mu)$; do not tighten grading merely because more precision is needed. The first $K_\mu \geq 8\kappa$ reaches a certified patch in $\text{poly}(I) + O(\log \kappa + B_\gamma)$ stages. Allocating at phase q at most $2^{q-\mu}$ bit steps to each $2 \leq \mu \leq q$ gives finite overall work $f_a(p, \kappa) \text{poly}(I + B_\gamma)$.

The output is a fixed integer assignment, a selected continuous face, and a rational strongly convex polynomial subproblem whose unique minimizer is an original global optimizer. It is an exact implicit description, not a rational-coordinate claim. Full search scheduling, integer-label fixation, and patch-certificate details are proved in the boundary companion. The margin term is material: existing examples have exponentially large B_γ at fixed degree, width and conditioning. Weakly complementary boundary cases without another certificate remain outside this termination theorem.

7 Initial implementation and historical evidence

This section preserves the first-release measurements and their original source snapshots. The later implementation and its separately saved runs are described in Section 8; the figures below are not reruns of the updated solver.

The original pruned-grid diagnostic implemented two-pass tree messages, all coordinate min-marginals, geometric grids, interval filtering, and hull recentering. Its saved exact-rational record covers nine fixtures, 70 stages, 26,589 bag-table states, and 526 pruned intervals. Separate finite enumeration checked min-marginals on 21,034 assignments. These figures describe that diagnostic, not the continuation’s benchmark suite. It did not implement the unknown-growth caps, rational reconstruction, or an independent stored-history verifier. Its successes establish finite regression evidence, not competitive general solver performance.

The continuation distinguishes three empirical questions: whether the produced certificate independently checks, whether filtering reduces state counts against comparable unfiltered methods, and whether its total time and memory are useful against other solvers. An ordinary floating-point solver result is useful as a timing baseline but is not automatically an exact global certificate. Controlled synthetic fixtures and public-library instances must be reported separately, with explicit limits and incomplete runs preserved.

7.1 The initial sparse solver and independent replay

The initial standard-library Python reference implementation handled supplied general tree decompositions, branching trees, empty separators, continuous and native-integer domains, fixed coordinates, and disconnected factor graphs. It implements corrected geometric grids, all coordinate min-marginals, domain filtering, and the capped unknown-growth approximation schedule. Coordinate-specific diagonal corrections sharpen the common- L bound. Endpoint reduction, bounded exact coordinate descent, and an optional rational PSD/KKT presolve supplement the grid algorithm. A saved result contains original-model bounds and a complete history, including when a resource limit interrupts refinement.

The separate verifier does not call grid generation, dynamic programming, filtering, coordinate descent, or the convex presolve. It shares rational model parsing and objective evaluation, independently reconstructs factor tables and Bellman equalities, checks all min-marginals and exclusions, and re-evaluates feasible incumbents. The certificate proves the embedded rational model; a caller must also match that model to the intended input. It is an independently implemented checker, not a formally verified one. Replay can cost as much as solving because it recomputes finite-table arithmetic. This first release returned certified approximations and certain exact presolve or finite-grid answers. General continuous rational reconstruction was a theorem at that stage; the completion described below adds its implementation.

7.2 Bounded comparison on original objectives

The main suite has ten small rational instances and two unmodified binary QPLIB instances. The small set includes random signed paths, bands and a branching tree, mixed and integer paths, flat and tied optima, and two positive-curvature scales. Random cases do not have planted optimizers; deliberate structural cases are labelled diagnostics. An independent exact integer-slice and active-face enumeration gives the small reference values.

All twelve cases use absolute tolerance $1/50$; a random path and both curvature cases also use $1/1000$. The resulting 15 configurations are run with four methods. Each has a two-second solver limit, with an external 14-second worker limit including setup, serialization and replay. Exact methods additionally have a 20,000 aggregate bag-state cap and 24-stage cap. SCIP is single-threaded with a 256 MB internal limit. The grid methods share a greedy minimum-degree decomposition; SCIP does not receive it. These timings come from one bounded run on a shared machine, not a statistical performance study.

Method	Small targets reached	Public targets reached	Incomplete runs
Geometric, pruned	13/13	0/2	2 table limits
Geometric, unpruned	13/13	0/2	2 table limits
Uniform, unpruned	11/13	0/2	4 table limits
SCIP (numerical bounds)	13/13	0/2	2 time limits

“Target reached” means the requested objective tolerance, not exact optimization. The SCIP small outcomes comprise six optimal statuses and seven gap-limit statuses. Its dual bounds remain numerical; its repaired box-feasible incumbents are separately evaluated exactly. All 45 rational grid certificates, including incomplete runs, pass replay. All 39 small-instance grid enclosures contain their exact reference values.

For the 13 matched small configurations, the pruned geometric method processes 6,167 cumulative bag-table states versus 27,386 without pruning. The strongest reduction in this small set is the high-positive-curvature case at tolerance $1/1000$: 771 versus 13,527 states. Tied-and-flat and pure-integer cases have equal counts for the two geometric variants. Thus pruning helps particular fixtures, while the suite does not show universal pruning benefit. Median solve times on these 13 configurations are approximately 0.0059 seconds with pruning and 0.0082 without it; median replay times are 0.0084 and 0.0078 seconds, respectively. Median whole-worker subprocess times are 0.174 and 0.193 seconds. Interpreter startup, setup, exact arithmetic and checking materially exceed some solve times. Peak worker memory, including replay, is about 25 MB on the small grid runs; it is not isolated DP memory.

The public inputs are QPLIB_3852 (231 binary variables, 440 quadratic terms) and QPLIB_5881 (120 variables, 2,123 terms). Their supplied decomposition widths are 25 and 95, respectively, and are heuristic upper bounds on treewidth. Each exact variant refuses its oversized tables before allocation and retains a valid initial bound; SCIP reaches its time limit. No public instance reaches the target. These outcomes demonstrate a practical width barrier for the supplied representations; they do not prove that better decompositions or formulations cannot work. The benchmark validates QPLIB’s triangular coefficient convention against the bundled library solutions and preserves the original objectives and domains, apart from sign reversal of a maximization objective.

7.3 Dimension and certified affine reduction

Fourteen additional runs test increasing path dimension, a width-two case, and the affine-recourse reduction. Across both first-release suites, all **56 rational certificates** pass independent replay;

the four affine enclosures also contain their analytic zero optimum. The main and extension suites use 20.23 and 22.23 seconds of summed subprocess wall time, respectively. No worker crashes or reaches its external wall limit. The following timings keep solve and replay separate.

Pruned random path	Gap	Solve (s)	Replay (s)	Peak MiB
$n = 16$	0.01497	0.0253	0.0231	24.3
$n = 64$	0.01361	0.2020	0.1967	24.8
$n = 256$	0.05786	2.0023	1.2388	46.5

The target is $1/50$. The first two paths finish; the 256-variable pruned path hits its time limit. Its unpruned counterpart hits a table limit at the same gap and uses 1.790 seconds for replay. SCIP reaches a numerical gap of approximately 0.000612 in 1.797 seconds on that path. A pruned width-two, 16-variable case also reaches its target, with gap 0.01953, solve time 0.0318 and replay time 0.0546 seconds. Three paths and one wider example do not constitute an empirical complexity law, and width one does not ensure fast Python rational arithmetic at every dimension.

The affine experiment has six original variables, four retained variables, and an exactly checked identity $F(u, v, y) = q(u, v) + M \sum_i (y_i - u_i)^2$ with feasible response $y = u$. It is the companion’s designed two-block example. The same reduced rational core is solved once for all three stiffness values.

Representation	States	Solve (s)	Replay (s)	Gap
Original, $M = 1$	1,694	0.0352	0.0425	0.000458
Original, $M = 100$	64,564	2.0044	2.0668	0.399817
Original, $M = 1000$	76,986	2.0033	2.5630	7.406978
Reduced, all M	394	0.0121	0.0151	0.000519

The target is $1/1000$. The stiff original cases time out; the reduced case finishes and its rational incumbent and lower bound lift exactly to each original instance. State counts are cumulative over completed stages. This demonstrates a useful certified preprocessing operation on the stated class. It does not show that these analytically simple models are difficult for other solvers, and the affine-selector recognizer itself is not benchmarked here.

7.4 Initial extension implementations

The first-release conditional-cut implementation included residual sign recognition, an exact rational max-flow oracle, primal/flow certificates, adaptive core cells, trace verification, and a rational-separation exact mode. Its broader concave/convex submodular extension had small exact diagnostics. The TU implementation was a bounded two-bag diagnostic; affine recourse and optimal-set/boundary extensions also had exact algebraic checks. General convex-QP factor evaluation, diagonal-class LP recovery, and reusable constrained-tree search were implementation gaps at that stage. Section 8 records which gaps the next release closes and states the remaining limits.

The saved experiment files provide per-run times, peak process memory, decomposition metadata, source hashes, rational reference values and replay certificates. The report’s coverage map points to them. The evidence supports a reusable certifying reference solver and concrete extensions; it does not establish competitive general-purpose MINLP performance.

8 Completed implementation and new evidence

The second implementation stage addresses the first release’s engineering gaps. Its source and benchmark records are separate from the historical snapshots in Section 7. The mathematical distinction between validity, termination and practical cost remains essential: an accepted rational certificate is exact, whereas a resource-limited search can finish without finding one.

8.1 Shared finite tables and automatic decompositions

The reusable finite-tree engine computes both message directions, all coordinate min-marginals and attaining assignments for finite factor tables, including infeasible entries. It checks tree structure and running intersection; the model adapters check factor coverage. Its callers build different objective and feasibility tables but share this computational rule. Certificate checkers separately validate the model-specific mathematical bounds. The base and constrained checkers independently implement Bellman replay; the convex-value checker reuses the finite-tree engine after independently checking its local optimization witnesses.

When a box-QP decomposition is omitted, a deterministic minimum-fill heuristic constructs one from the interaction graph. A supplied decomposition remains supported and is checked. The resolved bags and tree are serialized, so certificate replay does not depend on reproducing the heuristic. Returned width is an upper bound, not a minimum-treewidth claim. Table budgets are checked before oversized allocations. Sparse neighbor lists reduce zero-product work in coordinate polishing and gradients; dense model storage and exact rational arithmetic remain practical costs.

8.2 Exact rational output in the general grid solver

The `solve_exact` wrapper repeatedly requests finer original-model grid bounds, generates rational candidates, and applies Proposition 5.4. The height bound is computed once from the original coefficients and bounds. The implementation sharpens (19) using a product of integer Hessian row norms, which bounds every relevant principal determinant by Hadamard’s inequality. Neither filtered hull endpoints nor candidate denominators alter that original optimum-height bound.

Candidates include the current feasible incumbent, bounded-denominator coordinate reconstruction, and exact stationarity on a proposed active face. Singular stationary systems use rational linear feasibility. These are candidate-generation procedures: stationarity or proximity alone never establishes global optimality. The separate exact-output checker replays the underlying original-domain certificate, verifies the candidate against the original model, recomputes its objective and reduced denominator, and checks the strict rational-separation inequality.

The wrapper exposes round, stage, table and time limits, with configurable initial precision. An interrupted run retains its certified enclosure and says that exact output was not reached. Theorem 5.5 now proves eventual exact output for arbitrary nonunique rational box QPs with sufficiently expanded resource caps. The stronger point-growth running-time theorem retains the uniqueness hypothesis stated in Section 5. General finite termination does not provide that stronger efficiency guarantee.

8.3 Reusable exact linear and convex quadratic optimization

The shared rational optimization module implements two-phase simplex with Bland’s pivot rule. Successful LP results carry primal/dual witnesses, infeasibility carries a Farkas witness, and unboundedness carries a feasible point and improving ray. The result checker verifies the original

rational constraints and algebraic witness without replaying simplex pivots. A pivot limit is inconclusive; it is not an infeasibility certificate.

The convex box-QP oracle checks positive semidefiniteness exactly and searches active faces with exact linear algebra and LP feasibility for singular faces. A feasible point with checked KKT signs certifies the global minimum of the convex quadratic. Face and pivot limits bound this reference implementation. Neither simplex nor active-face enumeration is claimed here as a polynomial-time implementation of the theoretical convex-QP oracle. Their purpose is to make the restricted algorithms executable while retaining independently checkable answers.

8.4 Original-model recourse composition

The recourse pipeline recognizes globally affine convex responses, substitutes accepted responses, solves the reduced model, and lifts the result to the original coordinates. Each accepted response has a checked positive-semidefinite private Hessian, feasible whole-box affine range, and whole-box KKT identities and signs. Thus the original and reduced optimum values coincide. The final certificate stores every reduction, including fixed-coordinate substitutions, so a checker can reconstruct the chain and verify the final incumbent in the original model.

Candidate private blocks can be supplied or proposed from continuous components, private leaf-bag variables and single coordinates. This selection is heuristic. A rejected block or exhausted recognition budget remains in the model, and a failure to discover a reduction does not prove that no useful partition exists. The reduced problem can use the sparse grid solver or signed-concave minimum-cut recourse. The latter adapter checks signs and separates a small core before applying the existing exact conditional-cut search. The full reduction and search cost belongs in the reported total; preprocessing is not a free input promise.

Changing-active-set convex recourse is also executable. Supplied disjoint continuous PSD blocks, or heuristically discovered blocks, become local conditional value factors. The implementation caches rational convex-QP values and KKT witnesses at attachment grid points, uses corrected geometric grids on the retained coordinates, and supports exact original-QP height acceptance. Private feasible sets in this implementation are boxes; the general private-polytope theorem has a wider input class. The checker verifies local PSD/KKT data, recomputes finite-tree values using the shared engine, and checks the original-domain history. It invokes no LP or QP optimizer.

For scalar attachments, the integrated optional constructor enumerates private active patterns within a stated budget, certifies a complete interval partition, and uses its maximal private reduced curvature to sharpen the retained diagonal. Contributions from multiple scalar blocks add without a global overlay. The current constructor requires a positive-definite private Hessian and nonfixed intervals; unsupported or over-budget blocks retain the safe direct bound. The default pattern cap is 81. In tested changing-response examples this changes a retained bound from $2M + 6$ to 6 for stiffness $M = 1$ and $M = 100$. General multidimensional short-partition construction remains outside the implementation.

The mixed convex/concave submodular backend goes beyond minimum cuts. After verified sign changes, endpoint reduction of separately concave coordinates leaves a submodular set function evaluated by conditional convex QPs. Rational LP cutting planes minimize its Lovász extension; greedy chains supply separating affine functions. The final lower bound uses a finite convex mixture of checked greedy base vectors, and a feasible conditional minimizer attains it. Its independent checker verifies the conditional KKT witnesses and the mixture bound. This is an exact capped reference implementation, without a polynomial iteration guarantee for its cutting-plane search. It stores completed cached query witnesses, so its certificate storage follows the query budget; the implementation does not claim the theorem’s polynomial certificate-size bound.

8.5 Coupled constraints and optimal sets

The reusable TU solver uses feasible common-mesh tables on general tree decompositions, all coordinate min-marginals, and complete filtering history. Constraint scopes join objective scopes in the decomposition. The input includes an exactly checked TU certificate: supported structural forms include signed incidence and consecutive-ones matrices; a bounded all-minors fallback handles small matrices. A refused TU verification is not a proof of non-TU structure. Original integral alignment, finite integer labels and verified continuous curvature remain assumptions. The optional union representation preserves separated surviving cells. Exact output combines original-face rational candidates, the generic stationary-face recovery LP and a height-separation check. With resource caps removed, the recovery construction also covers arbitrary nonunique quadratic minima; only the finite-projection subclass has the stronger state bound. The width-dependent exponent and initial capacity costs stated in Section 6 are unchanged.

The optimal-set module constructs a compact Bellman-residual descriptor for the entire optimal set of a coordinatewise concave mixed-box QP. A separate checker verifies the recurrence and the support conditions, including possible interior integer labels. For the diagonal-certifiable continuous class, the implementation actually runs finite proximal trials with guessed growth ratios, performs rational stationary recovery, and accepts only the original KKT and positive-semidefinite certificate. A false growth guess cannot justify acceptance. Exhausting the configured search limits yields an inconclusive result, not a negative membership decision for the class.

8.6 Explicit polynomials and exact implicit boundary output

The polynomial implementation accepts explicit rational monomial factors, constructs or verifies their sparse decomposition, and computes upper coordinate-curvature bounds by exact interval arithmetic. It uses the shared finite-tree engine, corrected grids and complete pruning histories. Native integer output has the coefficient-denominator lattice stopping rule from Section 5.2, computed after substituting any fixed rational coordinates. Its checker independently recomputes that denominator and the strict pre-acceptance gap. Continuous approximation does not round an algebraic optimum to a rational exact answer.

The boundary search feeds finer globally certified polynomial boxes to a separate exact acceptance test. It first fixes native labels, then repeatedly certifies derivative signs and clamps the corresponding original bounds. Such a clamp preserves at least one global optimizer; a weak sign need not preserve every optimizer. On the remaining face, let H_0 be the midpoint Hessian and let δ bound the maximum absolute row sum of $H(x) - H_0$ throughout the box. An exact positive-definiteness check on $H_0 - \delta I$ proves uniform strong convexity. The polynomial together with this rational face box is then an exact finite descriptor of one global optimizer, possibly with irrational coordinates and value.

The boundary verifier independently recomputes monomial derivative intervals, reductions and the matrix bound, after replaying the global grid proof. It does not optimize the patch. The tested polynomial $(x^2 - 1/2)^2 + y + xy^2$ yields a certified patch for $(x, y) = (1/\sqrt{2}, 0)$ after four accuracy rounds. A bounded restart search implements discovery; it does not implement the theoretical bit-step dovetail or claim its FPT runtime. Failure of these sufficient tests, or a resource limit, can leave the result inconclusive, and general growth-only polynomial boundary output remains unresolved.

8.7 New bounded comparisons and certificate checks

The second-stage main suite has 73 configurations. Every solve runs sequentially in a fresh process with one numerical-library thread, a two-second cooperative solver budget, a five-second hard worker deadline, and a 512 MiB address-space cap. Replay runs in a separate five-second worker. The table cap is 30,000 states per stage. Saved timings separate solving, replay, preprocessing and total subprocess time; the latter includes both workers. Source snapshots and final hashes identify the implementation actually measured. Earlier provisional runs are archived and excluded from the following totals.

Lane	Configurations	Requests completed	Valid proof replays
Archived grid baseline	16	12	16
Completed grid	16	12	16
Exact rational output	9	9	9
Integrated recourse	11	10	11
TU constraints	16	13	15
Optimal-set descriptions	5	5	5
Total	73	61	72

Completion means the request made in that lane: the stated approximate gap, an exact value, infeasibility, or a checked optimal-set description. Partial bounds can pass replay without completing the request. The one configuration without a proof is an intentionally invalid TU input, correctly rejected before solving. Across the main suite, 54 enclosures contain independent exact reference values and 29 configurations certify exact values. The nine exact-output runs cover eight distinct models; a separable case repeated without convex presolve already has an exact initial bound, so that repeat is not evidence of nontrivial reconstruction. The fresh random path, band and mixed cases provide such evidence.

Both grid versions finish new random paths through 64 variables and a 32-variable random tree. They finish the same 12 of 16 requests, with summed solve-plus-replay subprocess times 15.18 and 15.34 seconds. This comparison jointly changes minimum-degree to minimum-fill decomposition, the message engine and sparse polishing; it establishes neither a speedup nor an isolated cause for a timing difference. Minimum fill reduces the supplied widths of the two full binary QPLIB instances from 25 to 19 and 95 to 93. Both still exceed the table budget, and neither public input reaches its target. The three continuous bound-only QPLIB entries screened from metadata have 14,400–39,204 variables; one has unbounded coordinates. They exceed the prototype’s declared 512-variable screen. These metadata refusals are not solver runs or successful continuous public benchmarks.

The integrated recourse tests operate on the original models. On a shuffled 17-variable affine star, automatic preprocessing removes 16 coordinates and certifies value -1 in 0.065 seconds of solve time and 0.047 seconds of replay. The 33-variable dense minimum-cut example reaches gap $1/1024$ in 0.089 seconds, with 0.099 seconds for replay. These are controlled structural examples. The two changing-response examples with stiffness $M = 1$ and $M = 100$ each use 17 table states, six levels and seven conditional QP queries, with certified gap $27/65536$. Their identical state counts illustrate this certified curvature reduction; two parameter values do not establish an empirical complexity law.

Two same-model constrained comparisons expose the value of the added structure. For $x_0(1 - x_0) + (x_1 - 1/3)^2 + (x_2 - 1/3)^2$ with $x_1 = x_2$, coordinate hulls hit the per-stage cap after 15 levels and 33,502 total processed states. Union filtering returns exact value zero after 43 levels and 2,538 total states, using two stationary-recovery LPs and 24 pivots; exhaustive face search is disabled. For

$1000(x_0 - x_1)^2 + (x_0 - 1/3)^2$ with $x_0 = x_1$, the ambient-curvature mode times out after 252,192 total states. The certified equality-tangent bound yields exact value zero with 1,938 states. These small examples exercise specific representation and curvature choices, rather than establishing general solver superiority.

Eleven separately frozen extension configurations exercise polynomial, implicit-boundary and mixed-submodular functionality. Seven reach their requested outcome; three retain checked intervals after a table limit, a cut limit or an unsupported signed class. One bounded symmetric-quartic boundary search remains inconclusive and emits no exact-output certificate. All ten emitted certificates replay. Eight numeric intervals enclose independent exact references, and two implicit patches contain independently known analytic optimizers.

The non-growth polynomial $(x - 1/3)^4$ reaches its requested gap in 36 stages and 545 states. A mixed polynomial needs eight stages and 469 states. The native-integer example, with one fixed rational coordinate, certifies value $-53/6$ after two stages and 17 states: its raw gap $1/24$ is strictly below its value-lattice spacing $1/18$. Independent enumeration checks all 100 native assignments. The unplanted six-variable signed mixed QP returns $-509/105$ using nine conditional QP queries, five greedy cuts and 88 pivots. A separate fixture requires a genuine mixture of two bases; a single-base lower bound $-13/16$ improves to exact value zero.

Across both new suites there are therefore **84 configurations, 82 valid proof replays and 68 completed requests**. Fourteen other runs retain checked incomplete or unsupported-class bounds. The invalid TU input and inconclusive boundary search are the two outcomes without certificates. Reference evidence comprises 62 numeric enclosures and two implicit-optimizer containment checks. The extension workers use the same hard time and memory limits and total 1.584 seconds of subprocess time; no process reaches a hard deadline or address-space cap. These figures exclude provisional archives and the unchanged 74 historical runs.

The final combined topic-specific test command passes 158 tests in 4.114 seconds. Its saved output and 35-source hash manifest accompany the completion package. These tests cover solver contracts, replay, invalid inputs, resource exits and certificate corruption. They are separate from the bounded benchmark configurations, and neither is a project-wide or CI verification claim.

9 Evidence map and interpretation

The corrected-grid proof, filtering-history argument, state bound, unknown-growth schedule, rational arithmetic, quadratic height bound, and exact-output construction appear in Sections 3–5. They develop the existing October 2 pruned-grid result; they are not claimed as newly discovered in this continuation. Section 6 includes the central structural and composition proofs. Longer imported arguments are identified in the companion coverage map. Sections 7 and 8 distinguish historical measurements from the completed implementation and its new runs.

Result	Established status	Boundary
Filtered mixed-box QP	Approximation and exact-output proofs; implemented rational recovery and independent original-model replay	Point-growth rate requires uniqueness; resource limits retain incomplete certified enclosures
Rational optimization	Reusable exact LP and convex box-QP with primal/dual, Farkas, ray or KKT witnesses	Simplex and active-face implementations do not have the theoretical polynomial oracle’s rate

Result	Established status	Boundary
Affine convex recourse	Exact reduction and growth transfer; automatic candidate discovery, recognition and lifted certificates	Discovery is heuristic; reduced fill and curvature must be controlled
Convex value factors	Exact composition and implemented conditional box-QP factors with changing active sets	General private polytopes remain theorem scope; direct retained curvature can remain large
Piecewise convex recourse	Reduced-piece curvature theorem without a global overlay; integrated certified scalar construction	Explicit partition size is part of theorem input; general short-partition discovery is not promised
Conditional minimum cuts	Automatic residual/core recognition; exact rational oracle, search, verifier and exact mode	Core size and sign-consistent separate concavity; no generic width-only recourse
Mixed submodular recourse	Implemented greedy-cut LP search, conditional QP witnesses and checked global mixture bound	Finite cut search and active-face QP can be exponential; stored proof size follows query budget
Explicit polynomial grids	Rational interval curvature, sparse global bounds and native-integer lattice exactness	Complexity theorem fixes numerical degree; continuous exact output requires another certificate
Integral TU fibers	Feasible-grid theorem and reusable tree solver; exact output with nonunique minima; union-state extension	XP width, initial capacity/alignment costs; finite-projection state bound excludes continuous optimal components
Optimal-set descriptions	Endpoint full-set constructor; actual unknown-growth search and acceptance for diagonal class	General efficient discovery for arbitrary unknown optimal sets remains unresolved
Polynomial active face	Implemented exact implicit output with checked face discovery	Termination theorem needs strict complementarity and an active-gradient precision cost
General sparse ν/g target	Scoped reductions and representation obstructions	No complete unrestricted algorithm
Practical solver value	Bounded comparisons and independent certificate replays, with incomplete runs preserved	High-width refusals and exact arithmetic costs prevent a broad competitive-performance claim

A sound certificate, a quantitative running-time theorem, and measured solver usefulness are different claims. Certificate validity follows from checked rational inequalities and algebra. Runtime bounds require their stated structural and growth assumptions and computational model. Practical usefulness requires suitable experiments. Independent research-agent review and finite checks do not constitute journal peer review or a proof of priority.

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